BRADFORD T. DHILLON

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Proactive and curious Wake Forest University graduate with a passion for the sciences, design, business, and technology. Natural leader with data analytics and business strategy experience who thrives in ambiguous environments that require bringing together people to solve technical and business challenges.

EDUCATION

Wake Forest University **Winston–Salem, NC**

B.S. in Engineering, Concentration in Biomedical Engineering *May 2021*

University of Sydney School of Engineering **Sydney, Australia**

International Foundation of Study Abroad *August 2019*

RELEVANT COURSEWORK

- Engineering (w/ labs): Materials and Mechanics, Measurements, Transport Phenomena, Computer Science, Control Systems & Instrumentation, Computational Modeling in Engineering
- Lab Sciences: Chemistry, Physics
- Mathematics: Calculus 1-3 (including multivariable), Linear Algebra, Discrete Math, Statistics

SKILLS / ACADEMIC EXPOSURE

- Software: PyCharm(Python), IntelliJ (Java), GitHub, MATLAB, Fusion 360, AutoCAD, MS Excel, Arduino Suite
- Hardware: PIXIS X-Ray Camera, Gel Electrophoresis, Tensile Tester, Electron Microscopy, 3D Printer, Oscilloscope, DMM, Soldering, Venturi meter, Spectrometer, Gardner Impact Tester
- Other: Numerical Methods, Reverse Engineering (Structural Hierarchies, Functional Modeling, System Benchmarking), Life Cycle Assessment & Modeling, Fractography

PROFESSIONAL EXPERIENCE

NASA- Goddard Space Flight Center **Greenbelt, MD**

Rapid Prototyping & 3D Modelling Intern *May 2017 – July 2017*

- Collaborated with material scientists to develop initial 3D rendering of X-Ray source bank for CMIST mission
- Independently designed and iterated on 3D printed hardware for PIXIS X-Ray Camera

ST Engineering- Middle River Aerostructure Systems **Middle River, MD**

Machine Calibration Intern *May 2019 – August 2019*

- Worked at General Electric turbofan engine plant calibrating all different machines on factory floor (torque wrenches, CNC mills, industrial autoclaves, industrial ovens, etc.)
- Calibrated various machines found in the engineering labs at the plant (laser cutters, tensile testing machines, etc.)

Zimmer Biomet **Washington, D.C.**

Technology Specialist *October 2021 – September 2023*

- Management and development of orthopedic robotics programs at +26 accounts across the Mid-Atlantic territory
- Carry out certification and onboarding of new physicians to be trained to use the ROSA orthopedic surgical robot as well as the PersonaIQ smart implant
- Work with orthopedic practices to integrate the MyMobility digital patient management platform into existing workflows

Program Development Associate II *October 2023 – Current*

- Manage existing and develop new robotics programs across multiple territories on the East Coast
- Work with physicians and product design team to iterate on existing technologies to problem solve and achieve surgical goals
- Work with physicians to use their intraoperative surgical data to draw conclusions about their practice

ACADEMIC PROJECTS

Designing a Ventilator – *Individual Project*

- Leveraged principles of functional modeling to create a plan for a new ventilator that could be mass produced more efficiently than existing designs in a state of emergency
- Evaluated efficacy of the lifetime of the materials used while being conscious of the environmental impacts of created waste
- Findings of the report concluded that most effective plan was to focus on changing several aspects of ventilator design that would make it more easily constructed by retooling existing assembly lines

Designing an Arteriovenous Stenosis Detection System– *Senior Capstone Team Project*

- Led a team effort focused on the research and development of a new technique for detecting arterial blockages in dialysis patients

- Coordinated communications with several outside technical experts including clinicians from dialysis clinics and biomedical engineers working in the industry
- Once accuracy of new technique was proven to be statistically significant on a fabricated test setup, a new product was designed to utilize it a way that could be easily adopted by clinicians at a dialysis clinic

ACTIVITIES AND COMMUNITY

Engineering Society @ University of Sydney, Hydrating Humanitate @ Wake Forest, Sigma Alpha Epsilon